

Asansol Engineering College

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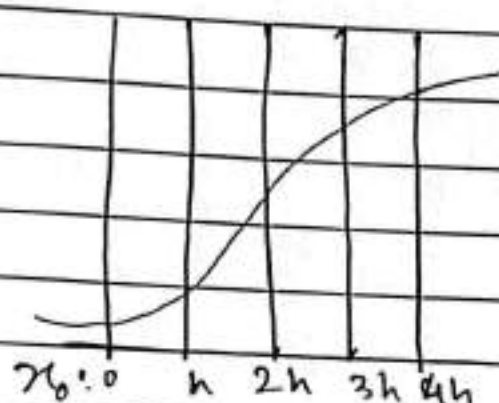
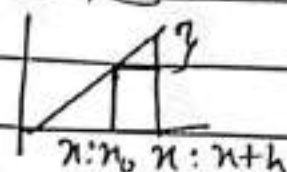
Year: 3rd

Subject: Numerical method

Sub. Code: OEE-IT 601A

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Numerical Analysis



finite differences

Grid $\rightarrow h$

Approximate value (Reasonable solution)

Δ forward diff operation

$$\Delta f(x) = f(x+h) - f(x) \quad \text{Current} = \Delta y_i = y_{i+1} - y_i$$

Next

$$\Delta \Delta f(x) = \Delta [f(x+h) - f(x)]$$

A linear operation $T(f+g) = T(f) + T(g)$.

$$\therefore \Delta^2 f(x) = \Delta f(x+h) - \Delta f(x)$$

$$= f(x+2h) - f(x+h) - f(x+h) + f(x)$$

$$= f(x+2h) - 2f(x+h) + f(x)$$

$$\nabla f(x) = f(x) - f(x-h)$$

Current - prev

$$\nabla^2 f(x) =$$

Teacher's Signature.....

Shift operation:

$$E f(n) = f(n+h)$$

$$E^2 f(n) = f(n+2h)$$

$$E^3 f(n) = f(n+3h)$$

$$E^4 f(n) = f(n+4h)$$

$$E^n f(n) = f(n+nh)$$

$$E^{-1} f(n) = f(n-h)$$

$$E^{1/2} f(n) = f(n + \frac{1}{2}h)$$

Only rational powers

$$\Delta f(n) = f(n+h) - f(n)$$

$$= E f(n) - f(n)$$

$$= (E - I) f(n)$$

$$\Rightarrow \Delta = E - I$$

$$\nabla f(n) = f(n) - f(n-h)$$

$$= f(n) - E^{-1} f(n)$$

$$= (I - E^{-1}) f(n)$$

$$\nabla = I - E^{-1}$$

$$\Delta \nabla f(n) = \Delta [f(n) - f(n-h)]$$

$$= f(n+h) - f(n) - f(n) + f(n-h)$$

$$= E f(n) - 2f(n) + E^{-1} f(n)$$

$$= (E + E^{-1} - 2) f(n)$$

$$\therefore \Delta \nabla = E + E^{-1} - 2$$

Teacher's Signature.....

Q. Find $\Delta^2 (x^2 - 3x + 5)$ at $x=3$ given $h=1$
 $(x^2+1)^2 - 3(x+1) + 5 = x^2 + 3x - 5$

$$x^2 \cdot x^2 + 2x + 1 - 3x - 3 + 5 = x^2 + 3x - 5$$

$$2x - 2 \mid x=3$$

$$x=2$$

$$\Delta(2x-2) = 2(x-1) - 2 - 2x + 2 \\ = 2$$

If $f(x)$ be a poly of degree n
 $\Delta^n f(x) = k$

$$f(x+h) = f(x) + h^1 f'(x) + \frac{h^2}{2!} f''(x) + \dots \\ = f(x) + h D f(x) + \frac{h^2}{2!} D^2 f(x) + \dots \\ = (1 + hD + \frac{h^2}{2!} D^2 + \frac{h^3}{3!} D^3 + \dots) f(x)$$

$$e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots$$

$$E f(x) = e^{hD} f(x)$$

$$E = e^{hD}$$

$$1 + D = e^{hD}$$

taking log

$$h \log(1+D) = D$$

$$D f(x) = h \log(1+D) f(x)$$

$$\log(1+D) = D - \frac{D^2}{2} + \frac{D^3}{3} - \frac{D^4}{4} + \dots$$

Teacher's Signature.....

$$y_h \left\{ \Delta - \Delta^2 + \Delta^3 - \dots \right\} f(u)$$

$$\Delta = f - 1$$

$$E = 1 + \Delta$$

$$E^n f(u) = f(u + nh) = (1 + \Delta)^n f$$

$$E^n = (1 + \Delta)^n = \sum_{k=0}^n n C_k \Delta^k = E^n$$

$$(1+n)^n = n C_0 + n C_1 + n C_2 + \dots + n C_n$$

$$= \sum_{k=0}^n n C_k$$

Ex: if $f(u) = 1$

$$f(1) = 3$$

$$f(2) = 8$$

$$f(3) = 13$$

find $f(u) = f(u + 6)$
 $\frac{1}{h} h$

$$(1 + \Delta)^6 f(u)$$

$$= 18$$